Degree and Variable Lower Bounds for Diophantine Definitions

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Abstract

Let K be a number field and let S be a finite set of its finite primes. A single polynomial whose K-rational input projection is an infinite subset of the ring of S-integers must have total degree at least four and at least three total coordinates. The coordinate bound also holds for finite systems of arbitrary degree, even with nonzero conditions and finite alternatives. For a single polynomial, the input and every essential witness coordinate must each have individual degree at least two. The proof combines an elementary argument about prime divisors of polynomial values, Siegel's theorem for integral points on curves, and Kollár's unirationality theorem for cubic hypersurfaces. Cubic cones and geometrically reducible components are treated explicitly.

1 Definitions and statements

For a nonzero prime ideal $\mathfrak{p}$ of the ring of integers $\mathcal{O}_K$, write $v_\mathfrak{p}$ for its additive valuation, normalized to have value group $\mathbb{Z}$, and put $v_\mathfrak{p}(0) = +\infty$. For a finite set S of nonzero prime ideals, define

$$\mathcal{O}_{K,S} = \{a \in K : v_\mathfrak{p}(a) \geq 0 \text{ for every } \mathfrak{p} \notin S\}.$$

In particular, $\mathcal{O}_{K,\varnothing} = \mathcal{O}_K$.

Definition 1.1. A single-polynomial presentation over K is an equation

$$P(T, Y_1, \ldots, Y_m) = 0, \qquad P \in K[T, Y_1, \ldots, Y_m].$$

Its *input projection* is the set

$$A_P = \{t \in K : \exists \mathbf{y} \in K^m, P(t, \mathbf{y}) = 0\}.$$

The distinguished input T is counted among the coordinates; thus there are $m + 1$ total coordinates and m existential witnesses. Degree means total degree, including T. A witness coordinate is *essential* if the polynomial depends on it.

A finite system uses the same coordinates for all its equations. We also consider finite unions of presentations consisting of equations and nonzero conditions. Geometrically, each such presentation is a Zariski open subset of a common zero set. All coordinates are evaluated in K, and no ordering conditions or universal quantifiers are allowed.

Theorem 1.2. *Let K be a number field and S a finite set of finite primes.*

1. *If a single-polynomial presentation has an infinite input projection contained in $\mathcal{O}_{K,S}$, then it has at least three total coordinates and total degree at least four.*

2. *The same lower bound of three total coordinates holds for finite systems of arbitrary degree, with nonzero conditions and finite alternatives permitted.*

3. *In the single-polynomial case, the individual degree in T and in each essential witness coordinate is at least two.*

For $K = \mathbb{Q}$ and $S = \varnothing$, this applies in particular to every polynomial definition of $\mathbb{Z}$ over $\mathbb{Q}$. The degree restriction in part (1) concerns one polynomial. A system of quadratic equations is a different presentation, and is not excluded by that restriction.

Two established geometric results enter the proof. Siegel's theorem gives finiteness of integral points on affine curves of positive genus, including S-integral points over a number field [2, Chapter 8]. Kollár's theorem gives unirationality of a noncone cubic hypersurface of dimension at least two over a perfect field as soon as it has a rational point [1, Theorem 2]. All number fields are perfect. The other ingredients are proved below.

2 Rational functions and primes outside a finite set

Lemma 2.1. *Let $g \in \mathcal{O}_K[Z]$ be nonconstant. For every finite set Σ of nonzero prime ideals of $\mathcal{O}_K$, there exist $a \in \mathcal{O}_K$ and $\mathfrak{p} \notin \Sigma$ such that*

$$g(a) \neq 0, \qquad v_{\mathfrak{p}}(g(a)) > 0.$$

Proof. After replacing $g(Z)$ by $g(Z+c)$ for a suitable $c \in \mathcal{O}_K$, we may suppose that $b = g(0) \neq 0$. Choose a nonzero rational integer M belonging to every ideal in Σ; take $M = 1$ if Σ is empty. Write $g(Z) = b + c_1 Z + \cdots + c_d Z^d$, where $d \geq 1$ and $c_d \neq 0$. Then

$$h(N) = \frac{g(bMN)}{b} = 1 + c_1 MN + c_2 bM^2 N^2 + \cdots + c_d b^{d-1} M^d N^d \in \mathcal{O}_K[N].$$

For every rational integer n and every $\mathfrak{q} \in \Sigma$, we have $h(n) \equiv 1 \pmod{\mathfrak{q}}$. The norm polynomial

$$\prod_{\sigma : K \hookrightarrow \mathbb{C}} h^{\sigma}(N)$$

lies in $\mathbb{Z}[N]$ and has degree $d[K : \mathbb{Q}]$: its leading coefficient is the nonzero norm of $c_d b^{d-1} M^d$. Its absolute value is therefore greater than one for all sufficiently large positive integers n. Such an $h(n)$ is nonzero and is not a unit of $\mathcal{O}_K$. A prime ideal dividing it lies outside Σ and also divides the nonzero value $g(bMn) = bh(n)$. Undoing the initial translation gives the required a. □

Lemma 2.2. *Let $r \in K(Z)$ be nonconstant, and let $E \subset K$ be finite. There is an $a \in K \setminus E$ at which r is defined and $r(a) \notin \mathcal{O}_{K,S}$.*

Proof. First suppose that r is a polynomial. Choose a prime ideal $\mathfrak{p} \notin S$ at which all its coefficients are integral and its leading coefficient is a unit. Only finitely many primes are excluded by these conditions. Choose $0 \neq \pi \in \mathfrak{p}$. For $a = \pi^{-n}$ and sufficiently large n, the leading term of $r(a)$ has strictly smaller $\mathfrak{p}$-adic valuation than all other terms. Its valuation is negative, so $v_{\mathfrak{p}}(r(a)) < 0$. The numbers π^{-n} are distinct, and hence all but finitely many avoid E.

Otherwise write $r = f/g$ with relatively prime polynomials $f, g \in \mathcal{O}_K[Z]$ and with g nonconstant. The polynomial Euclidean algorithm over K and clearing denominators give $A, B \in \mathcal{O}_K[Z]$ and $0 \neq c \in \mathcal{O}_K$ satisfying

$$Af + Bg = c.$$

Enlarge S to a finite set Σ containing the prime ideals dividing c or any nonzero value $g(e)$ with $e \in E \cap \mathcal{O}_K$. By Lemma 2.1, there are $a \in \mathcal{O}_K$ and $\mathfrak{p} \notin \Sigma$ with $g(a) \neq 0$ and $v_\mathfrak{p}(g(a)) > 0$. The displayed identity implies $v_\mathfrak{p}(f(a)) = 0$. Thus $r(a)$ has negative valuation at $\mathfrak{p} \notin S$. Moreover $a \notin E$: if $a = e$ were in $E \cap \mathcal{O}_K$, the prime divisor $\mathfrak{p}$ of its nonzero value $g(e)$ would have been included in Σ. □

An integral K-variety is called *K-unirational* if it is dominated by an affine space through a rational map defined over K.

Proposition 2.3. *Let X be a K-unirational integral variety, let $U \subseteq X$ be nonempty and open, and let $f : U \to \mathbb{A}^1_K$ be regular. If*

$$f(U(K)) \subseteq \mathcal{O}_{K,S},$$

then f is constant.

Proof. Pull back f through a dominant rational parametrization. If f is nonconstant, the result is a nonconstant rational function on affine space. Shrink its permitted domain to a nonempty principal open subset. In characteristic zero, one of its partial derivatives is nonzero. Since K is infinite, we can fix the other coordinates in K so that this derivative remains nonzero and none of the polynomials defining the permitted open set vanishes identically on the resulting line. The restriction is a nonconstant rational function $r \in K(Z)$, and the parametrization is permitted at every K-point of the line except a finite set. Lemma 2.2 then supplies a value outside $\mathcal{O}_{K,S}$, a contradiction. □

3 Integral-valued functions on curves

Lemma 3.1. *Let C be a smooth projective geometrically integral curve over K of positive genus, and let $f \in K(C)$ be nonconstant. There are only finitely many $P \in C(K)$ at which f is defined and $f(P) \in \mathcal{O}_{K,S}$.*

Proof. The function f defines a finite morphism $C \to \mathbb{P}^1_K$. Removing its poles gives an affine curve $D = f^{-1}(\mathbb{A}^1_K)$ finite over $\mathbb{A}^1_K$. Choose generators $z_1, \ldots, z_r$ of its coordinate ring over $K[f]$. Each z_i satisfies a monic equation over $K[f]$.

There is a finite set S' of finite primes, containing S, such that all coefficients in these finitely many equations belong to $\mathcal{O}_{K,S'}[f]$. If $P \in D(K)$ and $f(P) \in \mathcal{O}_{K,S}$, then each $z_i(P) \in K$ is integral over $\mathcal{O}_{K,S'}$. That ring is integrally closed, so all $z_i(P)$ belong to it. Consequently these points are S'-integral in a fixed affine embedding of D. Siegel's theorem gives their finiteness [2, Chapter 8]. Its positive-genus hypothesis holds for D, and at least one pole was removed because f is nonconstant. □

Proposition 3.2. *Let X be an algebraic set over K of dimension at most one, let $U \subseteq X$ be open, and let $f : U \to \mathbb{A}^1_K$ be regular. If $f(U(K)) \subseteq \mathcal{O}_{K,S}$, then $f(U(K))$ is finite.*

Proof. Pass to the finitely many irreducible components, omitting components disjoint from U. Zero-dimensional components contribute finitely many points. If an irreducible curve is not geometrically integral, every K-point lies in the intersection of distinct conjugate geometric

components. These intersections form a finite set, so this case also contributes only finitely many points.

For a geometrically integral curve, take its smooth projective normalization C. Away from finitely many points the normalization is an isomorphism, and f pulls back to a rational function on C. A constant function has finite image. If the genus is positive, Lemma 3.1 applies. If the genus is zero and $C(K)$ is empty, only the finitely many points omitted in passing to the normalization can contribute. If the genus is zero and $C(K)$ is nonempty, then $C \simeq \mathbb{P}^1_K$; Proposition 2.3 excludes a nonconstant function with the asserted integral-value property on the permitted open subset. This proves the result. □

4 Hypersurfaces of degree at most three

Theorem 4.1. *Let $X \subseteq \mathbb{P}^N_K$ be the zero set of a nonzero homogeneous polynomial of degree at most three. For every open $U \subseteq X$ and regular function $f : U \to \mathbb{A}^1_K$, the containment $f(U(K)) \subseteq \mathcal{O}_{K,S}$ implies that $f(U(K))$ is finite. The same conclusion holds for $X = \mathbb{P}^N_K$.*

Proof. Projective space is unirational, so that case follows from Proposition 2.3. Factor the equation of a hypersurface over K and treat each irreducible factor separately. Multiplicities do not change the rational point set.

Geometrically reducible components. An irreducible polynomial of degree two or three over a perfect field, if not geometrically irreducible, splits geometrically into conjugate linear factors. Indeed the geometric factors have a common degree and form one Galois orbit; that degree divides the prime number two or three. Every K-point belongs to all the conjugate hyperplanes. Their common intersection is a K-linear subspace, and Proposition 2.3 applies to the restriction of f. Linear components are treated by the same proposition.

Geometrically integral quadrics. A quadric with a smooth K-point is rational by projection from that point. If it has no smooth K-point, all its K-points lie in its singular locus, which is a linear subspace. Both cases give the required finite image.

Cubics that are not cones. Curves and zero-dimensional cases are covered by Proposition 3.2. A geometrically integral cubic of dimension at least two that is not a cone is K-unirational if it has a K-point, by Kollár's theorem [1, Theorem 2]. This theorem applies to singular cubics as well as smooth ones. If there is no K-point the image is empty; otherwise use Proposition 2.3.

The structure of a cubic cone. Write its cubic equation as F on a K-vector space V and put

$$W = \{w \in V : \partial_w F = 0 \text{ as a polynomial}\}.$$

This is the kernel of a K-linear map. In characteristic zero, F is constant along the directions in W, so it descends to a cubic form on V/W. The projective space $\mathbb{P}(W)$ is the maximal vertex. Choose a K-linear complement to W. It gives a geometrically integral cubic base B which is not a cone. Indeed, a vertex direction of the base would lift to an additional direction in the kernel defining W. That kernel commutes with extension of the ground field, so the same assertion holds geometrically. Its dimension is at least one, since a zero-dimensional cubic hypersurface is not geometrically integral.

Outside $\mathbb{P}(W)$, projection from the vertex gives a morphism to B. Each fiber over a K-point of B is an affine space of dimension $\dim W$. On an open subset of B this bundle is trivial, so the cone is birational to $B \times \mathbb{A}^{\dim W}$. These statements follow directly by writing a point of V as the sum of a vector in W and a vector in the chosen complement.

If $B(K)$ is empty, all rational points of the cone lie in the vertex. If $B(K)$ is finite, apply Proposition 2.3 on the permitted open part of each affine-space fiber, and also on the vertex. This gives a finite union of finite images. If $\dim B \geq 2$ and $B(K)$ is nonempty, Kollár's theorem makes B, and then the cone, unirational.

A cone over a curve with infinitely many rational points. The remaining case has $\dim B = 1$ and $B(K)$ infinite. Let C be the smooth projective normalization of B. Discarding finitely many base points, write the function f on an open subset of $C \times \mathbb{A}^{\dim W}$. If this rational function depends nontrivially on the affine-space coordinates, some partial derivative in those coordinates is nonzero. Expressing it as a quotient of polynomials with coefficients in $K(C)$ shows that, outside finitely many base points, the specialized function is nonconstant and its permitted fiber is nonempty. Choose a K-point of C outside those exceptions. Its fiber is a nonempty open subset of affine space, contradicting Proposition 2.3. Therefore f belongs to $K(C)$.

For every K-point of C outside a finite set, the permitted fiber is a nonempty open subset of affine space and has a K-point. Thus the descended function is defined and S-integral at all these base points. Proposition 3.2 implies that its image there is finite. Since a nonconstant rational function on a curve has finite fibers and $C(K)$ is infinite, the function must be constant. It is then constant as an element of the cone's function field, and hence wherever it is regular on that irreducible cone. This includes the points omitted from the chosen birational chart. □

5 Consequences for polynomial presentations

Proof of Theorem 1.2(1) and (2). First consider at most two total coordinates. For one system of equations, its zero set is either all of the ambient affine space or has dimension at most one. In the first case, the permitted nonempty open subset is unirational, and its input function cannot have an infinite S-integral image by Proposition 2.3. In the second case, Proposition 3.2 gives a finite input image. Empty permitted sets contribute nothing. Taking finitely many alternatives preserves finiteness. This proves the coordinate bound for systems and for single equations.

For a single nonconstant polynomial of total degree at most three, homogenize it and regard the distinguished input as a regular function on the usual affine chart of its projective zero set. Theorem 4.1 gives a finite input image, independently of the number of coordinates. A nonzero constant polynomial has empty projection. The zero polynomial has projection K, which is not contained in $\mathcal{O}_{K,S}$ because S is finite. These observations prove the degree bound. □

Proposition 5.1. *Suppose that $P \in K[T, Y_1, \dots, Y_m]$ has an infinite input projection contained in $\mathcal{O}_{K,S}$. Then*

$$\deg_T P \geq 2, \qquad \deg_{Y_j} P \geq 2 \quad \textit{for every essential witness } Y_j.$$

Proof. Suppose first that P is linear in an essential witness Y_j. Write

$$P = A(T, \mathbf{Y}')Y_j + B(T, \mathbf{Y}'), \qquad A \neq 0.$$

Choose constants $\mathbf{c} \in K^{m-1}$ so that $A(T, \mathbf{c})$ is a nonzero polynomial. Such a choice exists because K is infinite. For every t outside its finite root set, setting $Y_j = -B(t, \mathbf{c})/A(t, \mathbf{c})$ gives a rational witness. The projection is therefore cofinite in K. It cannot be contained in $\mathcal{O}_{K,S}$, whose complement is infinite: for a prime $\mathfrak{p} \notin S$ and $0 \neq \pi \in \mathfrak{p}$, all the distinct elements π^{-n} lie outside $\mathcal{O}_{K,S}$.

Suppose next that $\deg_T P \leq 1$, and write $P = A(\mathbf{Y})T + B(\mathbf{Y})$. If $A = 0$, the input projection is either empty or all of K. Otherwise on the nonempty open set $A \neq 0$ the rational function $-B/A$ takes only values in $\mathcal{O}_{K,S}$. Proposition 2.3 makes it constant. At any rational point with $A = B = 0$, every input would be admitted, which is impossible. Thus the projection has at most one element, again a contradiction. This proves Theorem 1.2(3). □

Corollary 5.2. *No finite system of linear equations, even with nonzero conditions and finite alternatives, has an infinite input projection contained in $\mathcal{O}_{K,S}$.*

Proof. Each nonempty common zero set of linear equations is an affine space over K. Apply Proposition 2.3 to its permitted open subset and take the finite union of the resulting finite images. □

Corollary 5.3. *Let $A \subseteq \mathcal{O}_{K,S}$ be infinite. If A has a single-polynomial presentation of degree four over K, then four is the least possible degree of any such presentation, regardless of how many witness coordinates are allowed. If A has a presentation by a system of quadratic equations, then two is the least possible maximum degree of the equations in a system.*

Proof. The first assertion follows from Theorem 1.2; the second follows from Corollary 5.2. □

The coordinate lower bound permits any number of equations and arbitrary degrees. In contrast, the degree-four obstruction concerns the geometry of one hypersurface. Neither statement asserts the existence of a presentation in three total coordinates.

References

[1] J. Kollár, Unirationality of cubic hypersurfaces, *Journal of the Institute of Mathematics of Jussieu* **1** (2002), 467–476, Theorem 2. https://arxiv.org/abs/math/0005146.

[2] S. Lang, *Fundamentals of Diophantine Geometry*, Springer, 1983, Chapter 8, "Siegel's Theorem and Integral Points", 188–224. https://doi.org/10.1007/978-1-4757-1810-2_8.